

Shelf Intelligence & Predictive Replenishment Agent

Secure multi-platform AI for predictive retail execution

Mobile + Desktop

API-secured mobile

POS + Inventory + Planogram

No camera dependency

The Retail Problem

Promo and high-demand items run out before teams can react, creating avoidable revenue leakage.

Out-of-stock revenue leakage

- Promo items sell faster than expected
- Empty shelf means missed basket conversion
- Lost sales are often visible only after the promo

Reactive store operations

- Staff depend on manual checks
- Replenishment happens after the issue
- Priority items compete with routine tasks

Static shelf planning

- Planograms remain fixed while demand changes
- Slow-moving SKUs occupy valuable space
- POS insights are not translated into action fast enough

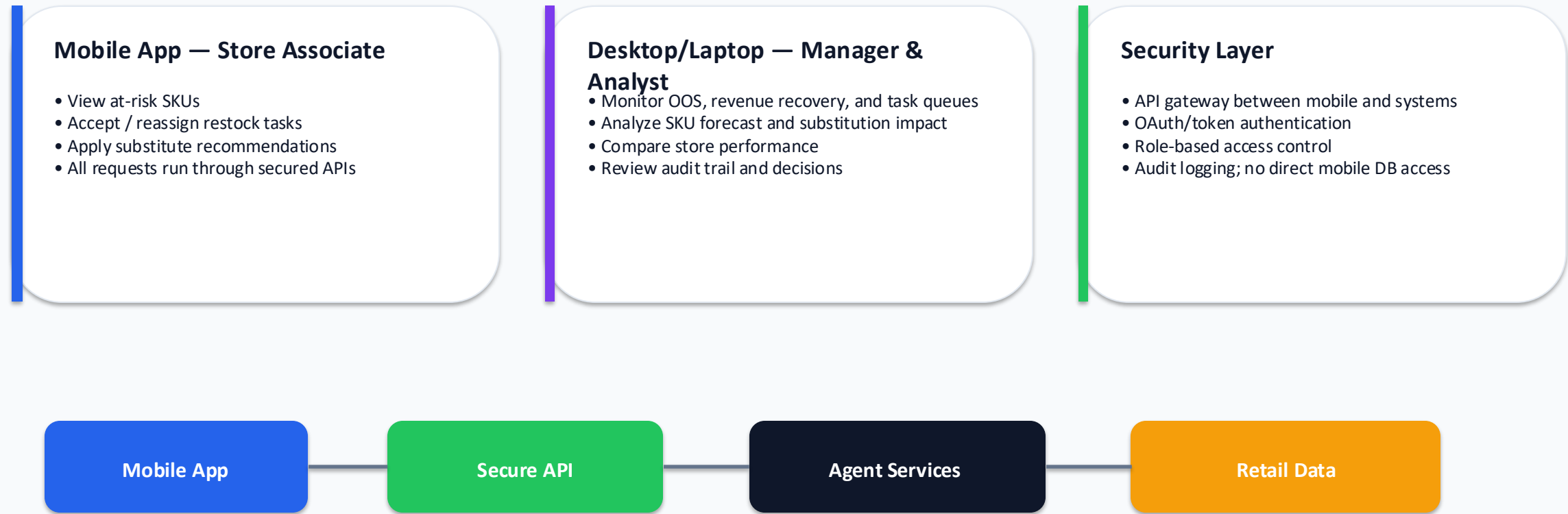
Solution Overview

A secure agentic workflow that predicts risk and recommends the best next action using existing retail data.



Multi-Platform + Secure-by-Design

Mobile is optimized for execution; desktop/laptop is optimized for monitoring, analysis, and management.



Security message for pitch: the mobile app never connects directly to POS, inventory, or planogram databases.

Agent Roles

A modular setup keeps the solution explainable, scalable, and easy to demo.

Demand Forecast Agent

Predicts SKU depletion using POS velocity, promo uplift, and historical patterns.

Replenishment Agent

Checks stock availability and recommends when and how much to replenish.

Recommendation Agent

Suggests substitute SKUs using similarity, margin, promo fit, and basket behavior.

Planogram Agent

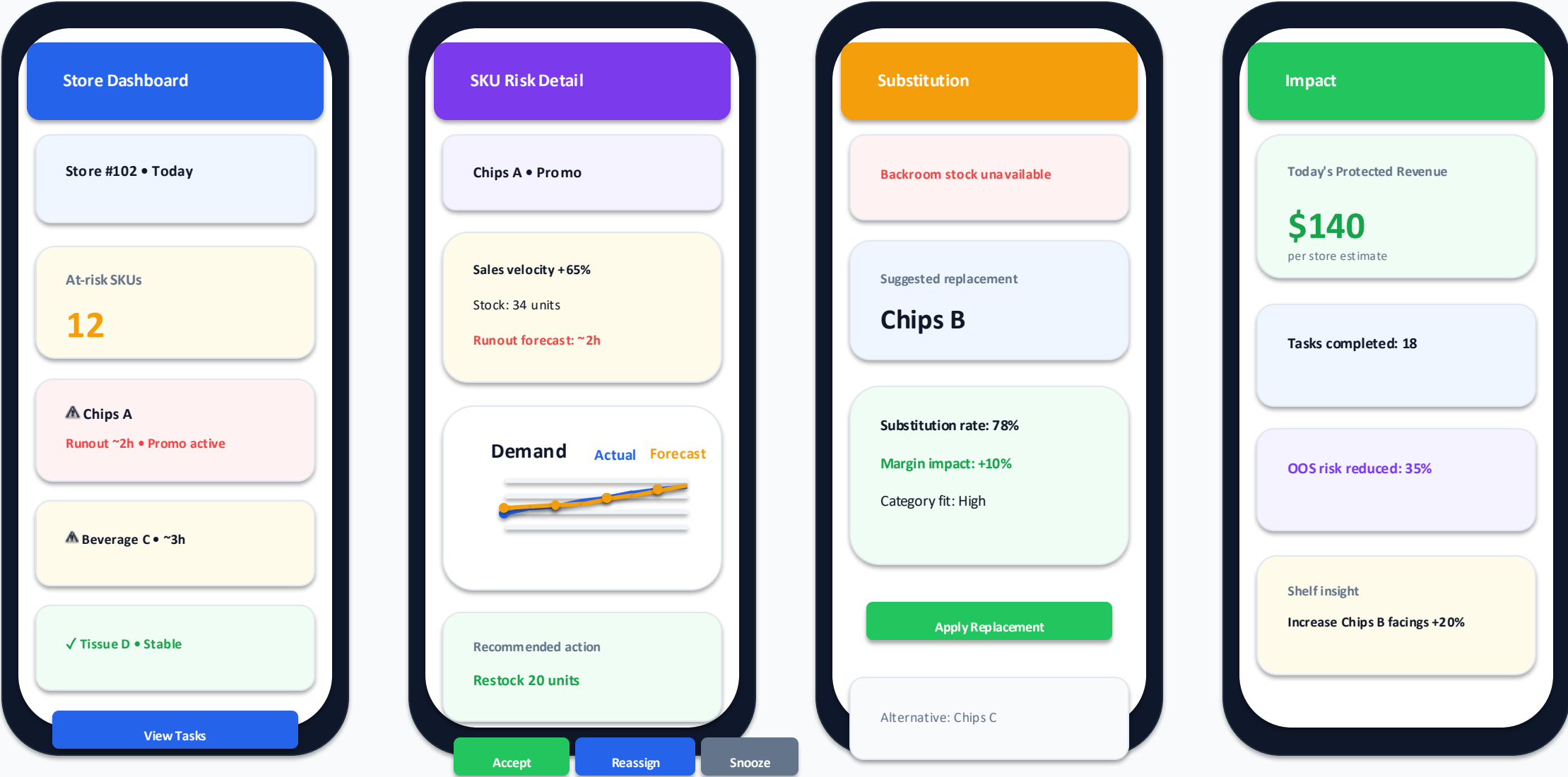
Compares allocated facings against sales performance and recommends shelf-space changes.

Task Orchestrator Agent

Prioritizes tasks and routes actions to store associates via mobile or desktop task queues.

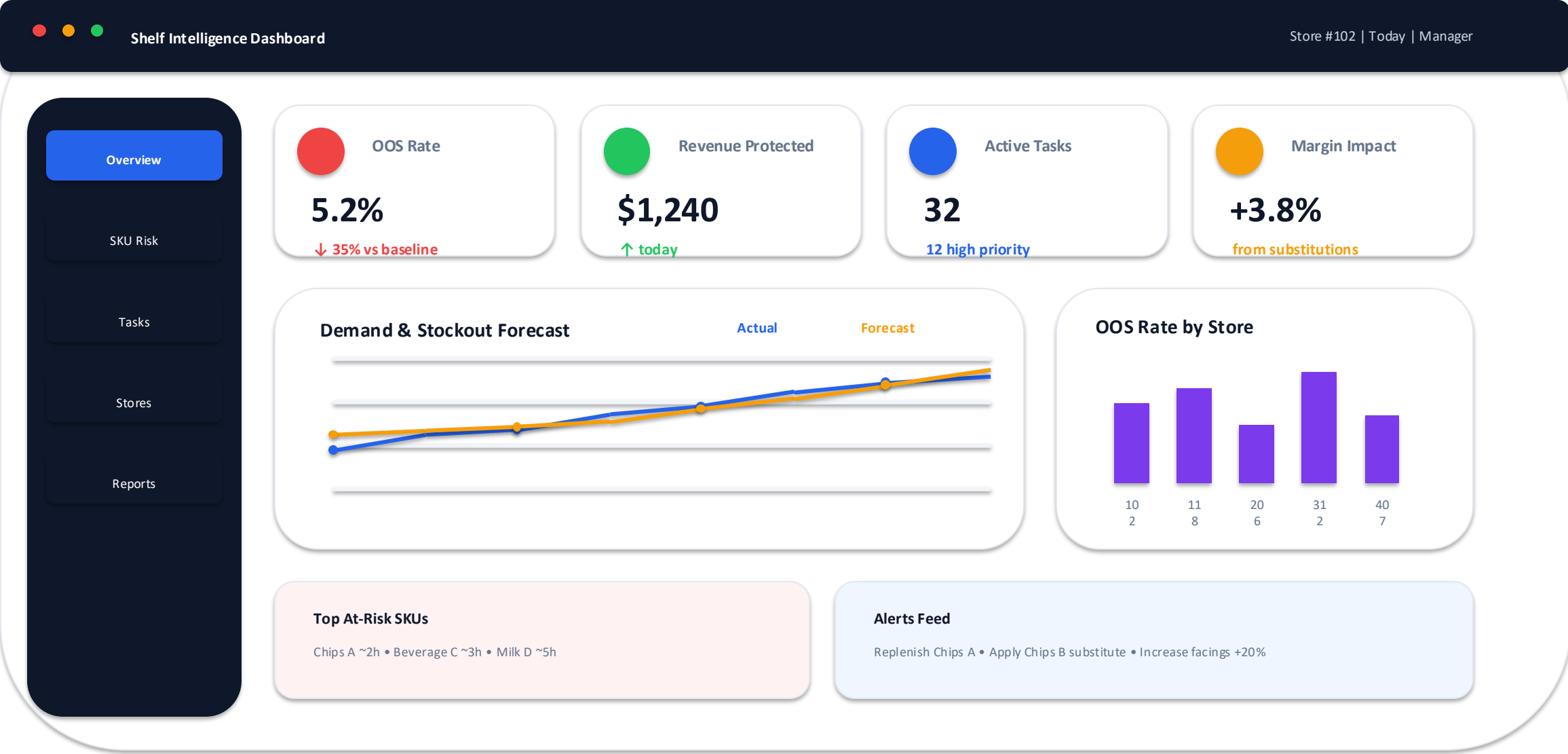
Concept Mobile UI Designs

Store associates get fast, action-oriented screens for replenishment and substitution decisions.



Concept Desktop UI Design — Manager Dashboard

Desktop/laptop experience for monitoring stockout risks, revenue recovery, and operational workload.



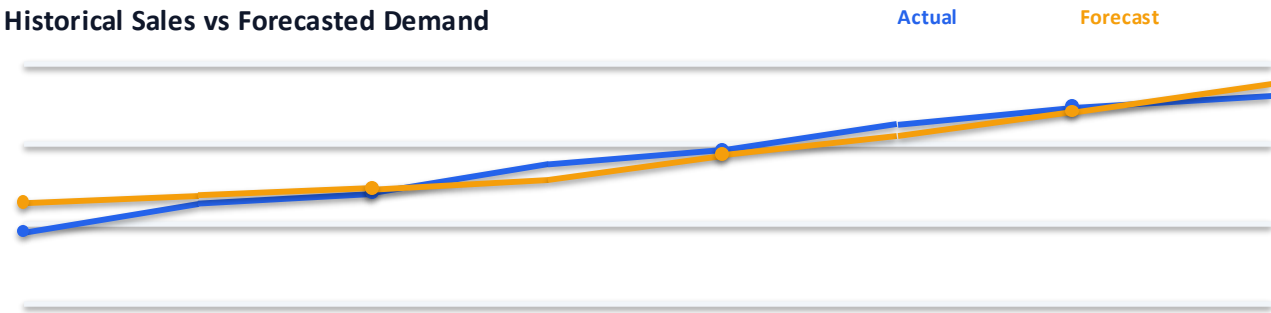
Concept Desktop UI Design — SKU Analysis

Deep-dive view for understanding demand, inventory position, recommended action, and substitution impact.

SKU Analysis: Chips A | Snacks | Promo Active

High Risk

Historical Sales vs Forecasted Demand



Recommended Action

Restock 20 units within 30 mins

If unavailable, apply substitute recommendation to protect conversion and margin.

Approve

Escalate

Stock Details

Current stock: 34 units
Backroom stock: 0
Runout forecast: ~2 hours

Substitution Recommendation

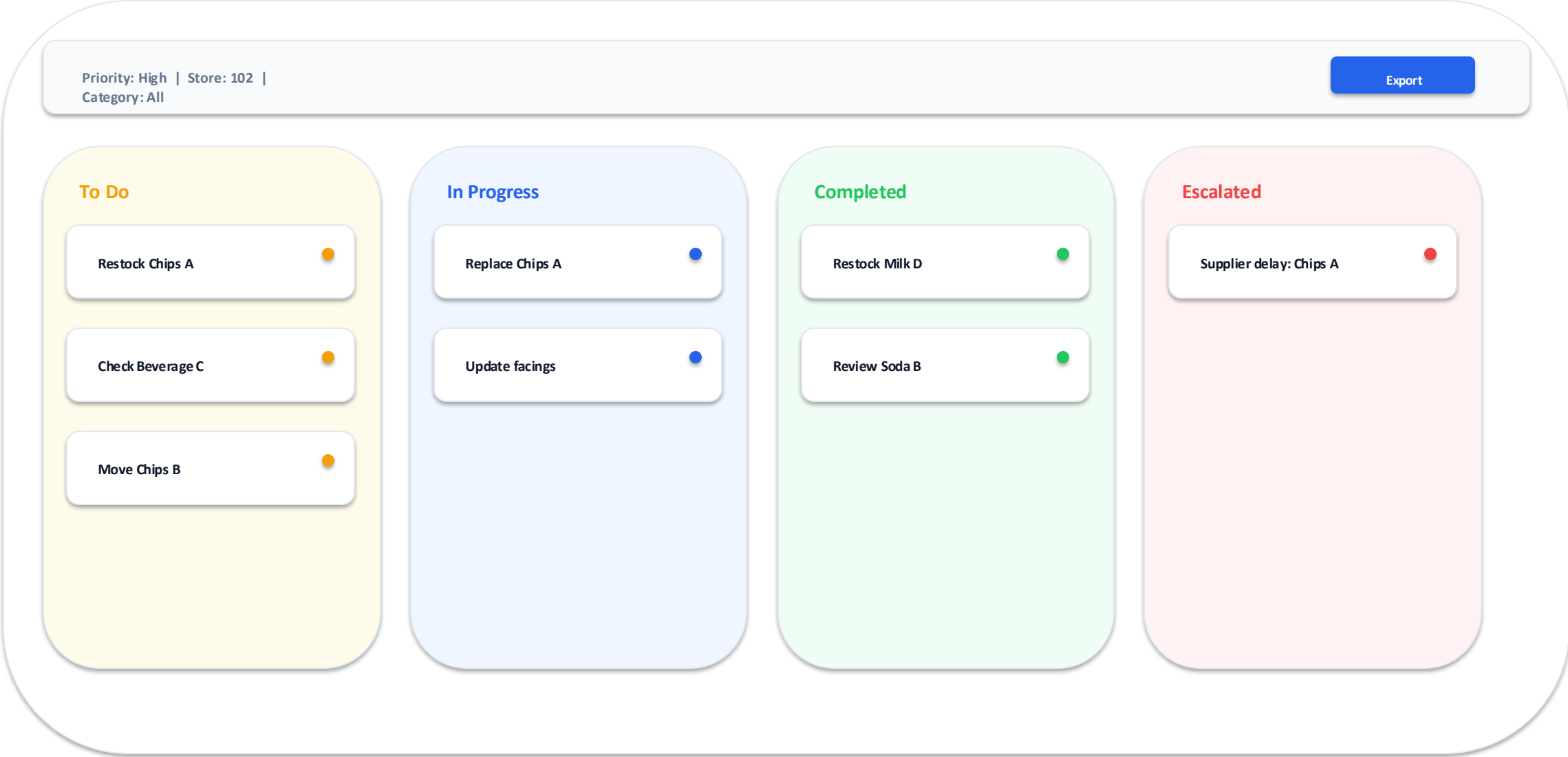
Replace with Chips B
Substitution rate: 78%
Margin impact: +10%

Planogram Insight

Increase Chips B facings by +20% temporarily while promo demand remains elevated.

Concept Desktop UI Design — Task Management

Kanban-style task board for tracking replenishment, substitution, and escalation work across the store.



Demo Script Storyboard

Use this sequence for a concise pitch demo showing prediction, action, substitution, and value.

1

Open Manager Dashboard

Show Store #102 with KPI cards, at-risk SKUs, and revenue protected.

2

Explain the alert

Chips A is promo-active and forecasted to run out in around two hours.

3

Switch to Mobile

Store associate receives a prioritized restock task.

4

Simulate no backroom stock

System switches from restock action to substitute recommendation.

5

Apply substitution

Agent recommends Chips B with substitution and margin impact.

6

Show impact

Manager dashboard reflects protected revenue, completed tasks, and OOS risk reduction.

Key Business Impacts

The pitch should emphasize direct revenue recovery plus operational efficiency.

Revenue Recovery

Recover sales that would otherwise be lost when promo or high-demand SKUs go out of stock.

Improved Promo Execution

Keep promotional items available during demand spikes and protect campaign ROI.

Higher Margin Decisions

Suggest substitutes that preserve conversion and may improve margin on affected baskets.

Labor Productivity

Reduce manual checking and guide staff to the highest-priority replenishment tasks.

Better Shelf Utilization

Move shelf space toward high-performing SKUs and away from slow movers.

Customer Experience

Reduce empty-shelf frustration and increase the chance customers complete their basket.

Monetary Impact Model

Illustrative model for pitching value; replace assumptions with actual retailer data when available.

Example Assumptions

Daily store revenue: \$10,000
OOS-related missed demand: 8%
Recoverable portion: 50%
Agent improvement: 35%

Lost sales due to OOS

$$\$10,000 \times 8\% = \$800/\text{day}$$

Recoverable lost sales

$$\$800 \times 50\% = \$400/\text{day}$$

Recovered by agent

$$\$400 \times 35\% = \$140/\text{day/store}$$

Annual value per store

$$\$140 \times 365 \approx \$51,100/\text{year}$$

100-store portfolio

$$\$51,100 \times 100 \approx \$5.1\text{M}/\text{year}$$

Note: Values are illustrative for demo purposes and should be calibrated with actual sales, inventory, and stockout data.

Reference Architecture

A practical implementation view using existing systems, secured APIs, and multi-platform channels.



Why This Wins

Fast to deploy

Uses existing POS, inventory, and planogram data — no camera or shelf hardware required.

Clear business value

Connects agent actions directly to recovered sales, margin protection, and labor productivity.

Enterprise-ready

Supports mobile + desktop usage with secured API access, RBAC, and audit logging.

From reactive shelf management to predictive, secure, data-driven retail execution.